



Lab Number & Title: Lab 01: Ubuntu Server Installation and Environment Setup

Student Name: Niba Sadiq

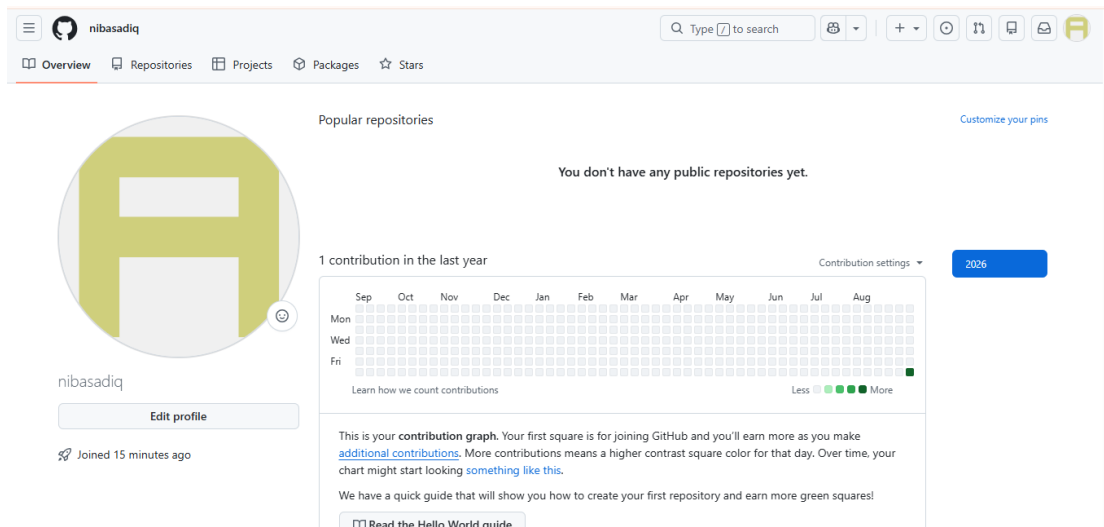
Registration Number: FA24-B1-SE-064

Course Code: SE-665 — Cloud Computing

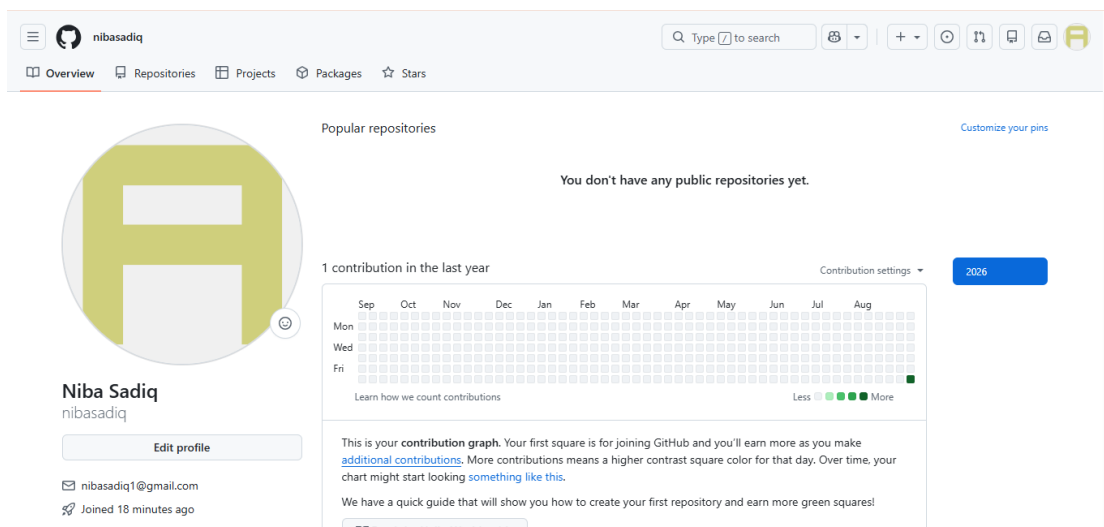
Section: Section A

GitHub Username: nibasadiq

Task 1: Create and Configure Your GitHub Account

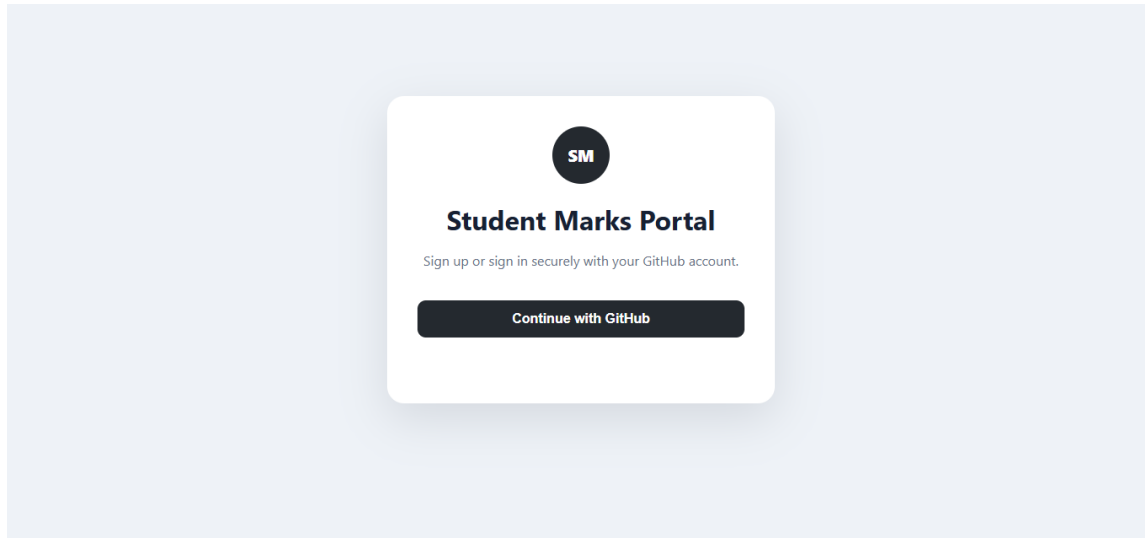


Profile overview showing the newly registered GitHub account



Shows the updated GitHub profile details including display name and contact email

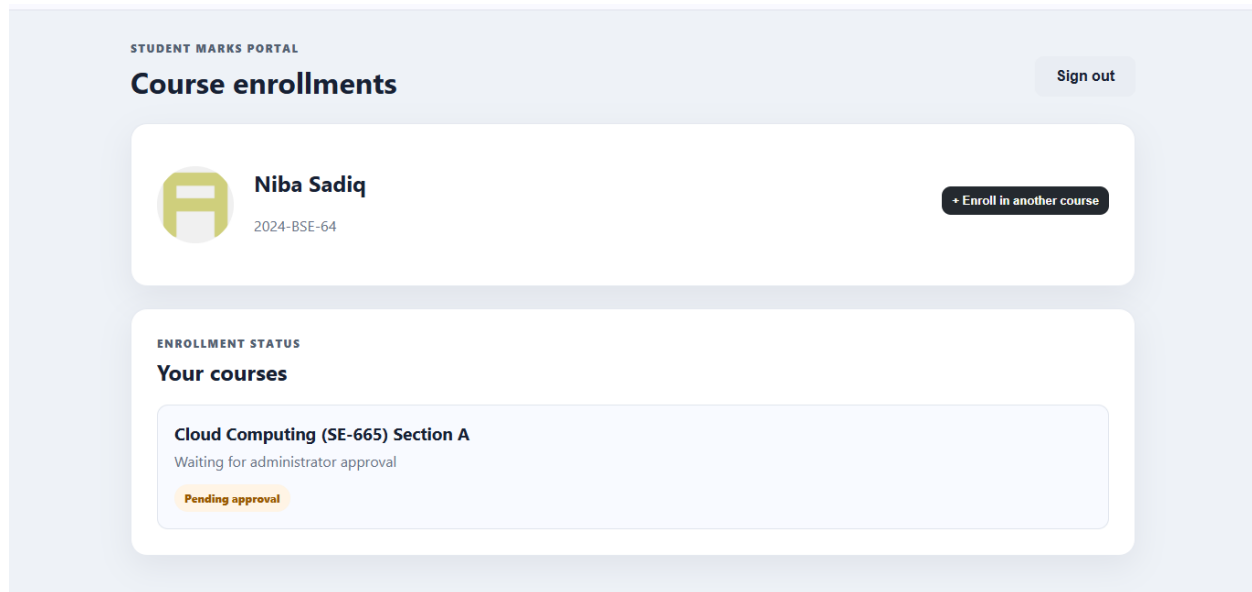
Task 2: Register in the Student Marks Portal



Sign-in landing page for the Student Marks Portal using GitHub OAuth.

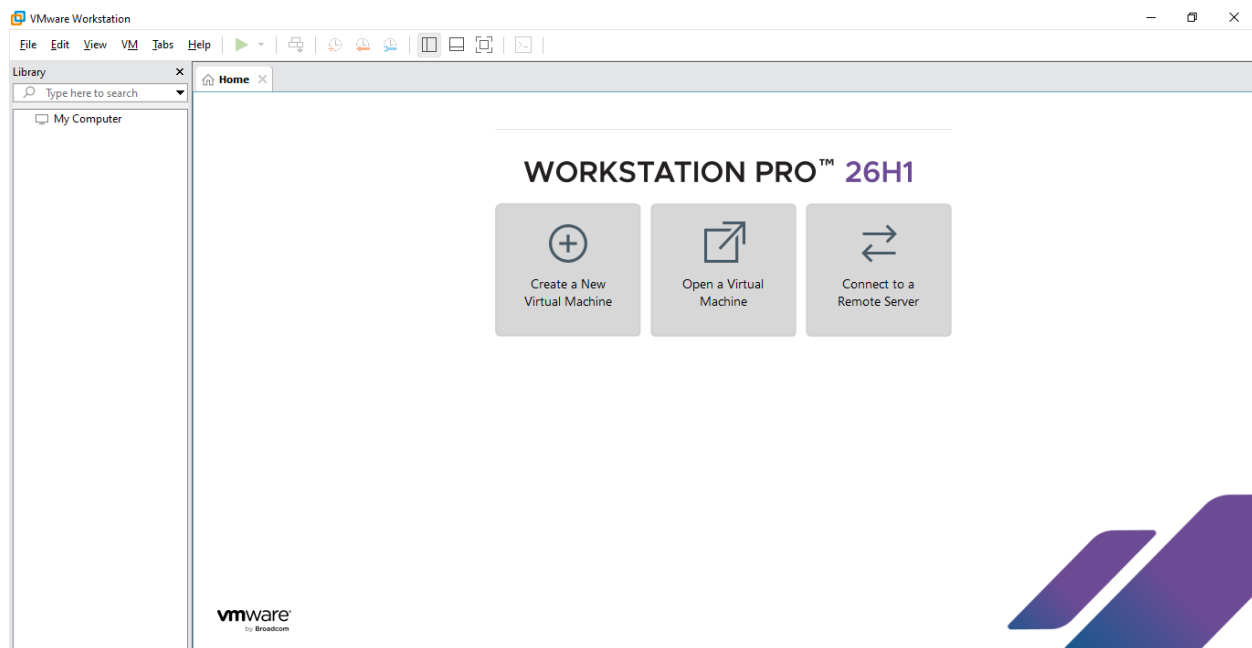
The image shows a 'Complete registration' form. It features a central white card on a light blue background. At the top of the card is a black circle with the letters 'ID' in white. Below this is the title 'Complete registration' in bold. Underneath the title is the text 'Enter your registration number, course and section. The enrollment requires approval.' The form contains three dropdown menus: 'Registration number' with the value '2024-BSE-64', 'Course' with the value 'SE-665 — Cloud Computing', and 'Section' with the value 'Section A'. Below these dropdowns is a dark blue button with the text 'Submit enrollment' in white. At the bottom of the card is a light blue button with the text 'Sign out' in dark blue.

Registration form filled with registration number 2024-BSE-64, course SE-665 — Cloud Computing, and Section A

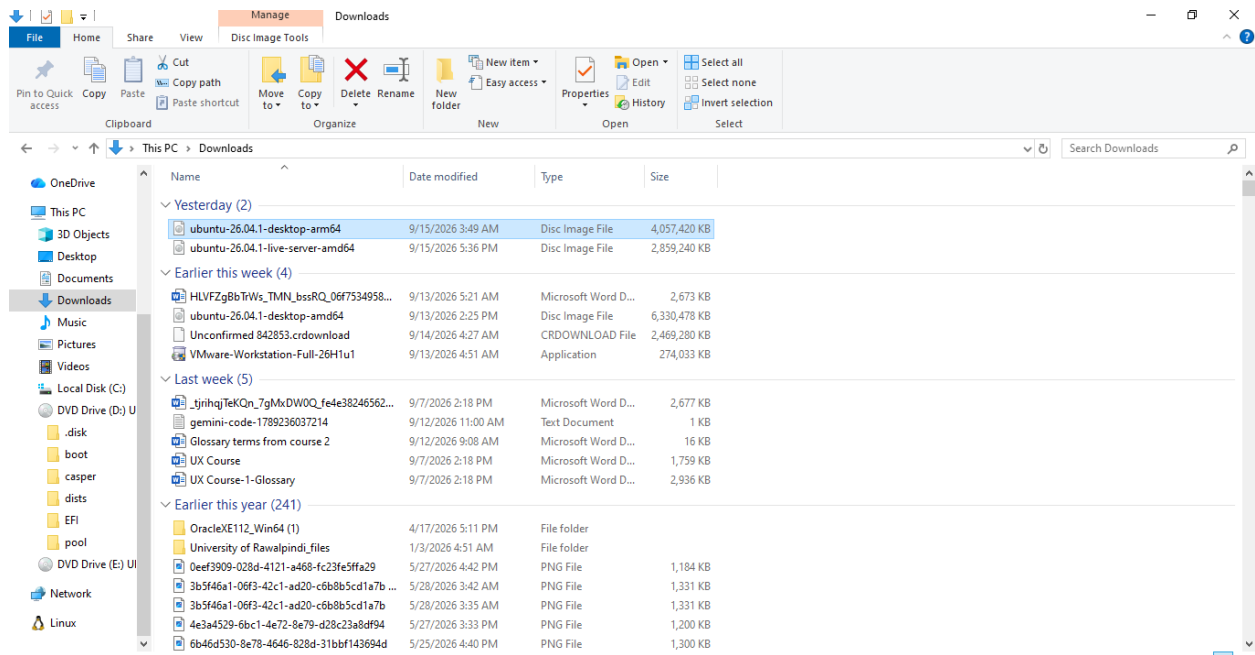


Shows pending course enrollment approval for Cloud Computing Section A.

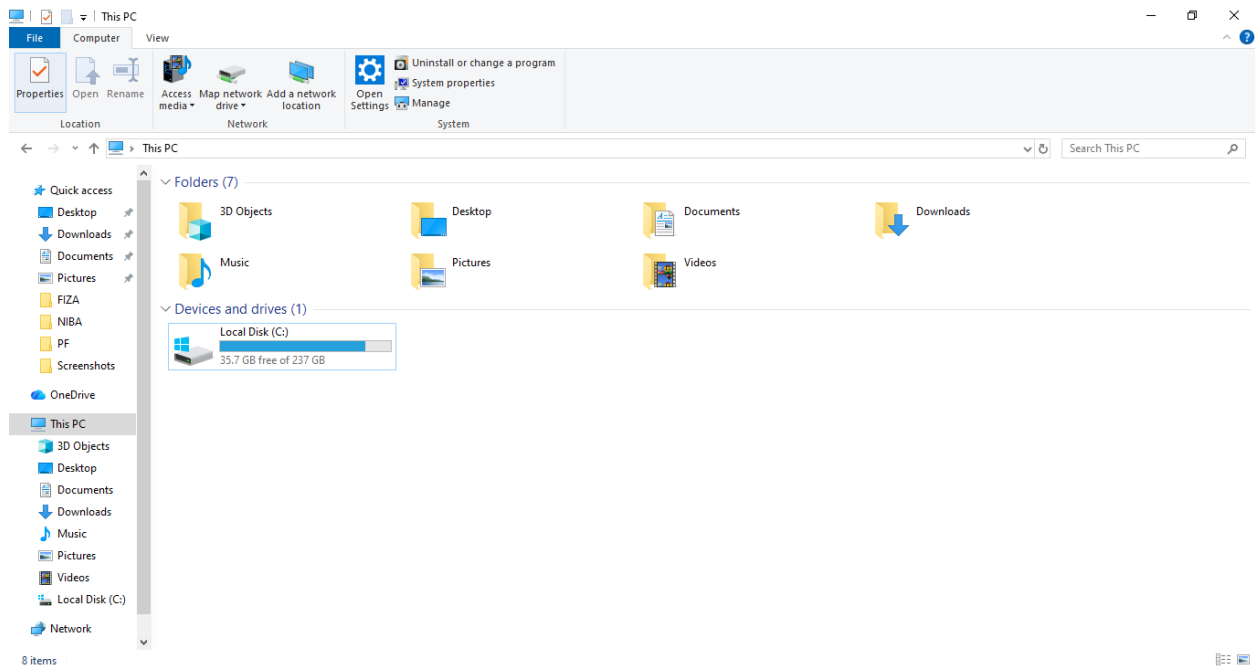
Task 3: Download VMware Workstation Pro and Ubuntu Server



VMware Workstation Pro home screen.

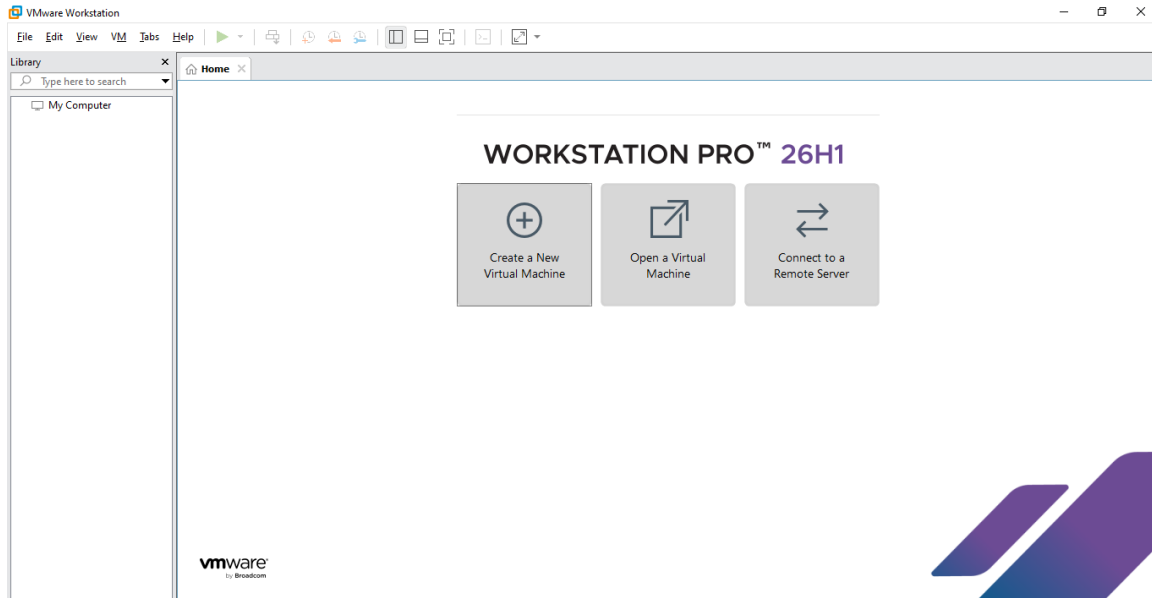


Downloaded Ubuntu 26.04 live server ISO file in local downloads directory.

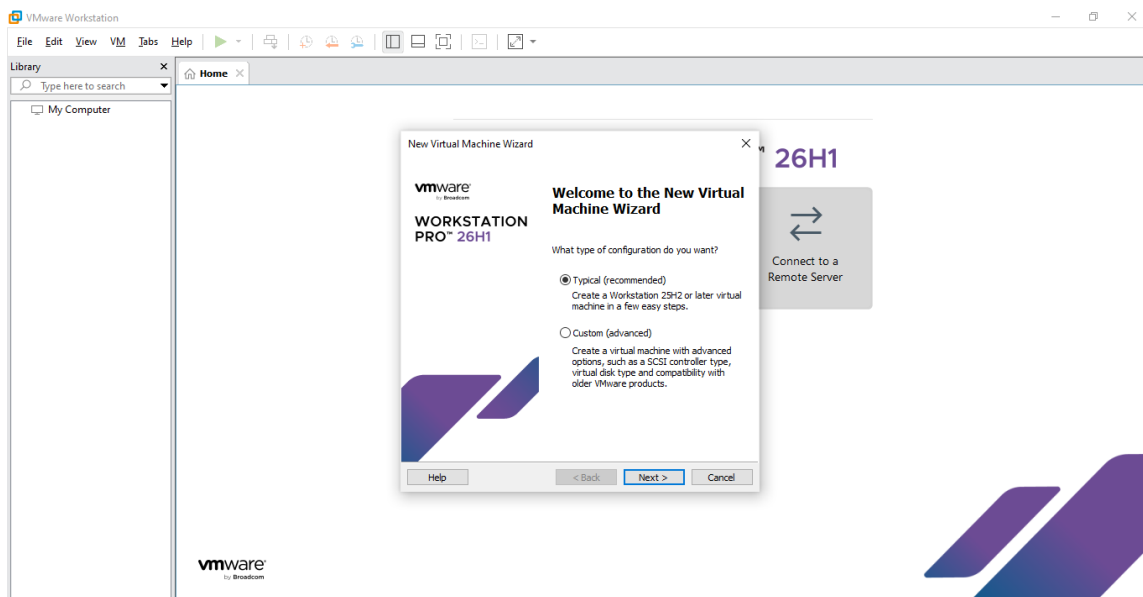


Verification of available disk storage for virtual machine installation.

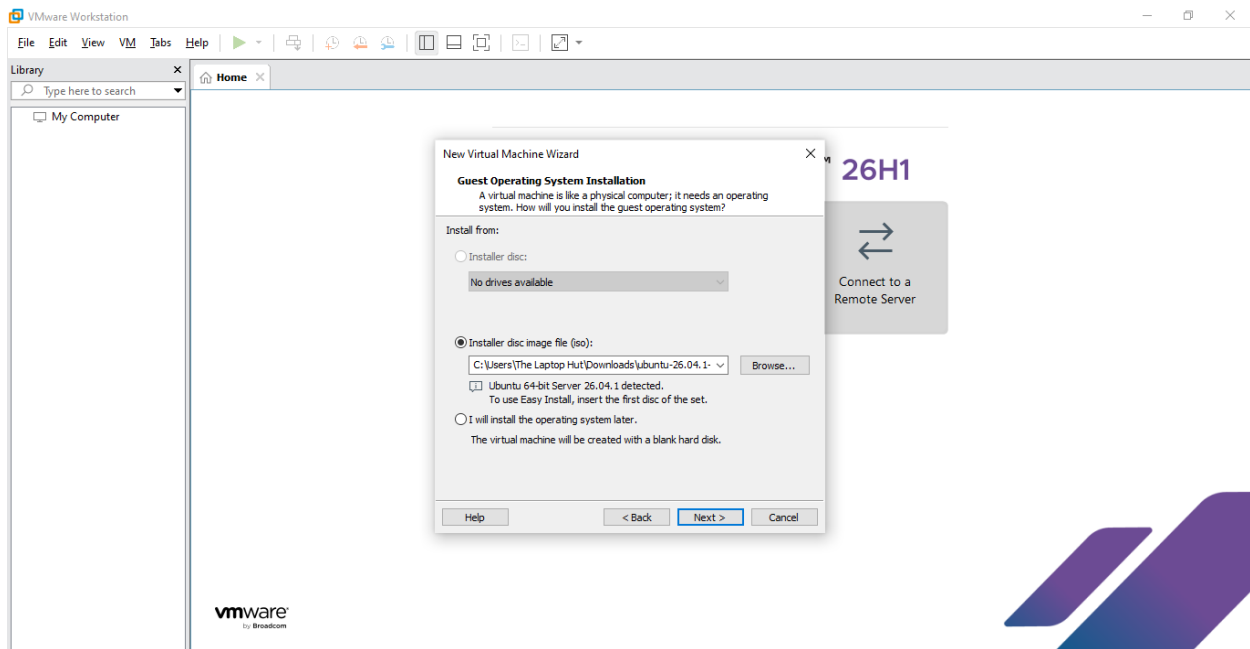
Task 4: Create the Ubuntu Server Virtual Machine



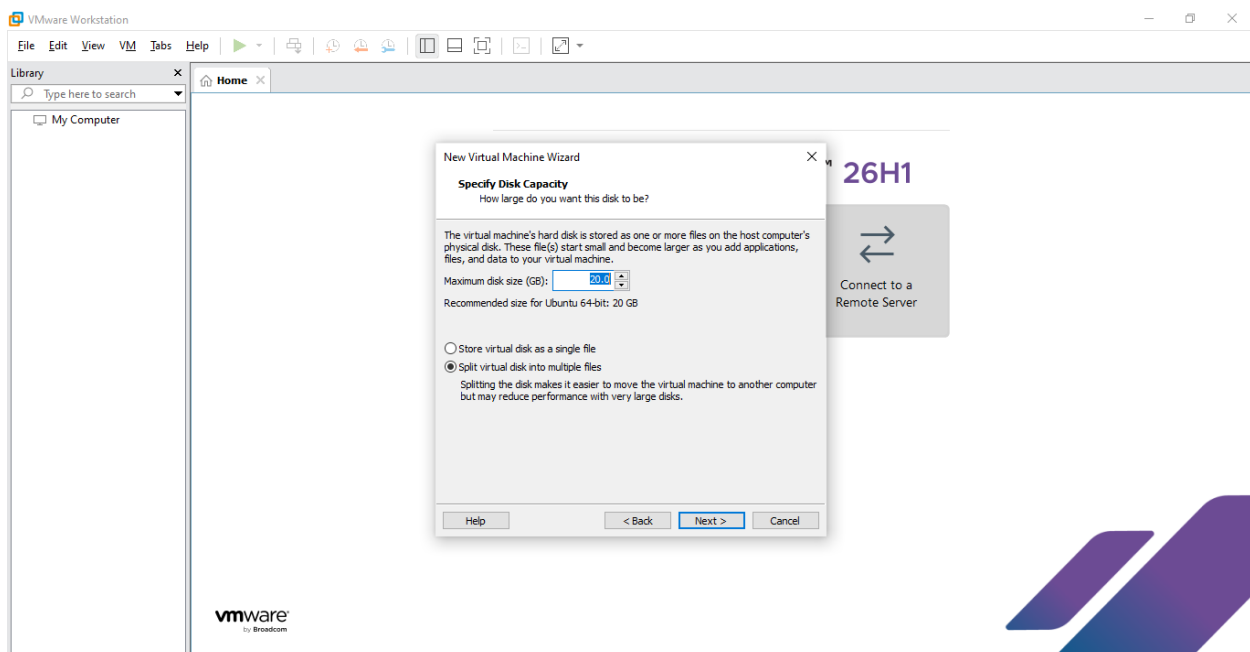
Selecting "Create a New Virtual Machine" in VMware Workstation.



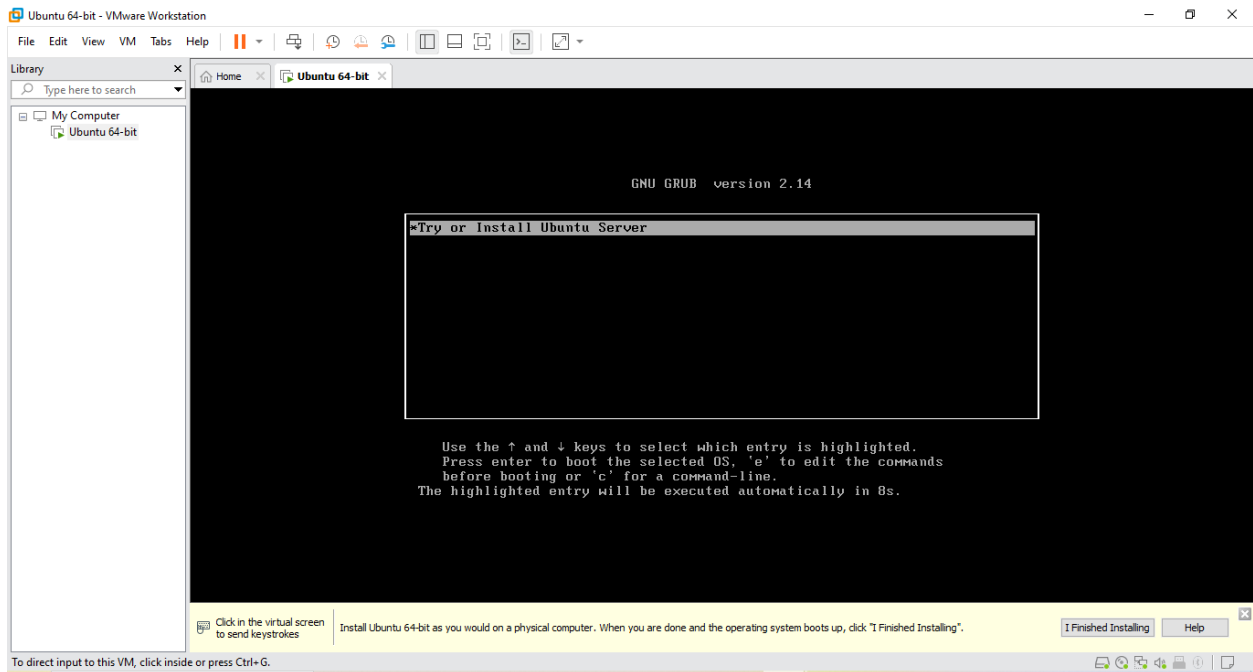
Selecting Typical configuration mode for virtual machine creation.



Browsing and selecting the target Ubuntu Server ISO image.

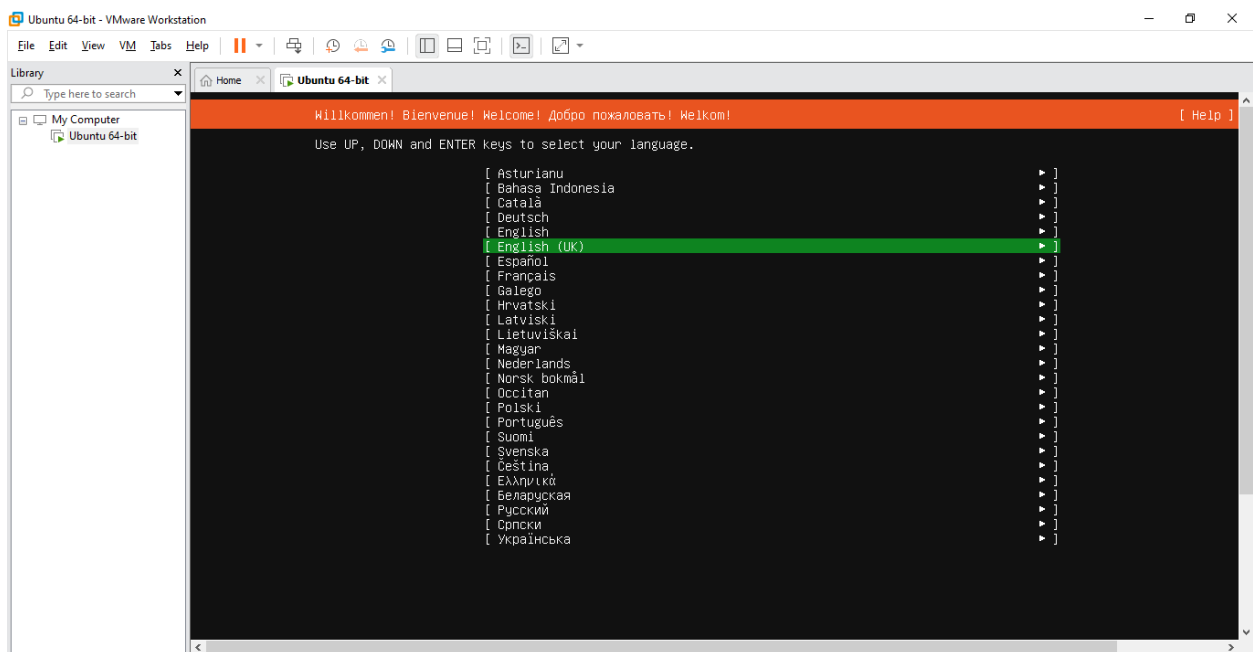


Setting virtual disk size to 20.0 GB and selecting split disk storage



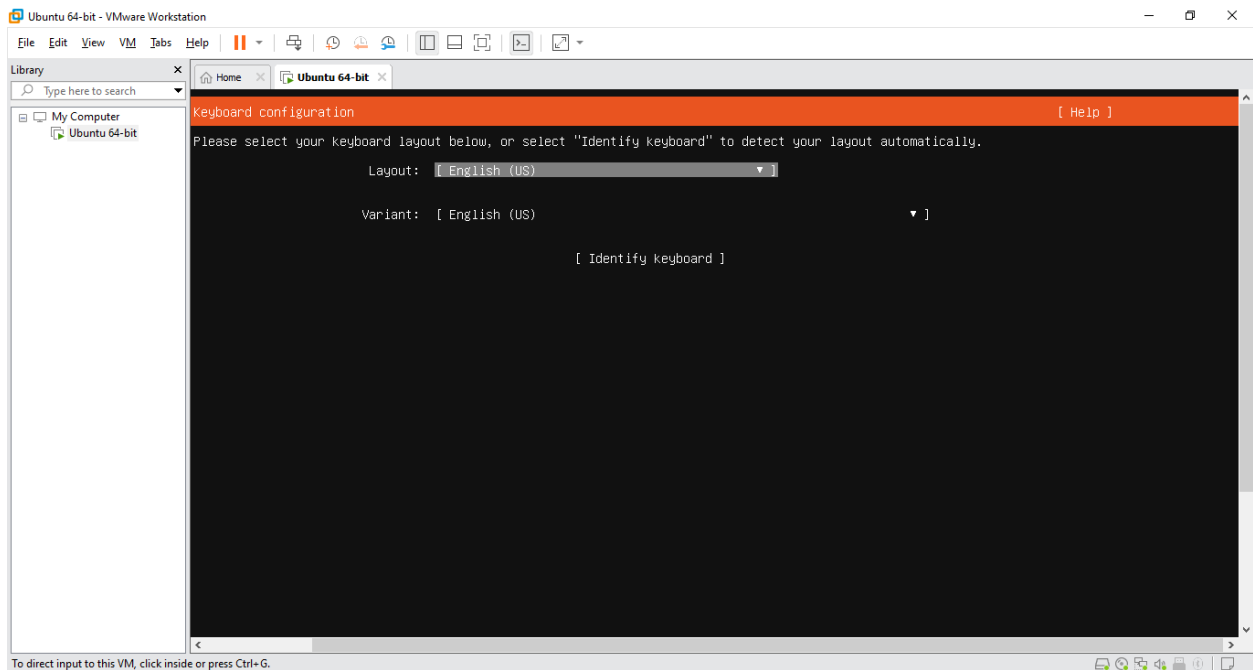
First boot of the virtual machine entering the Ubuntu installer GRUB menu.

Task 5: Choose the Installation Language

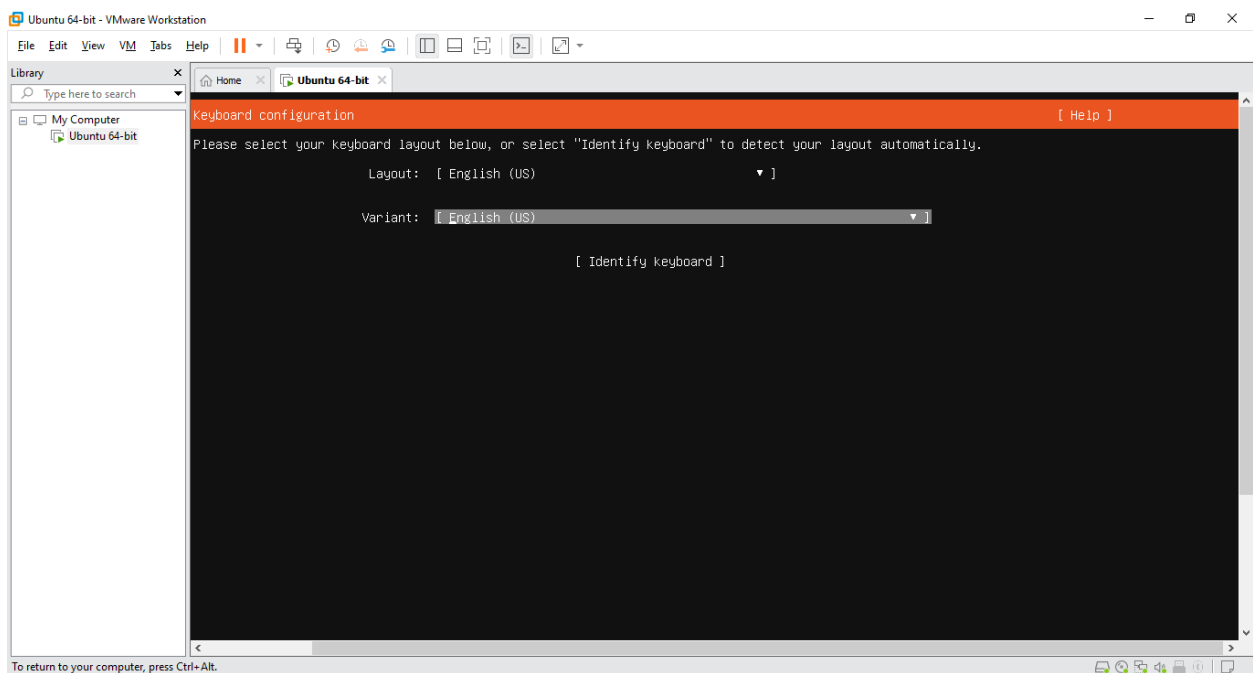


Language selection screen in the Ubuntu installer.

Task 6: Configure the Keyboard Layout

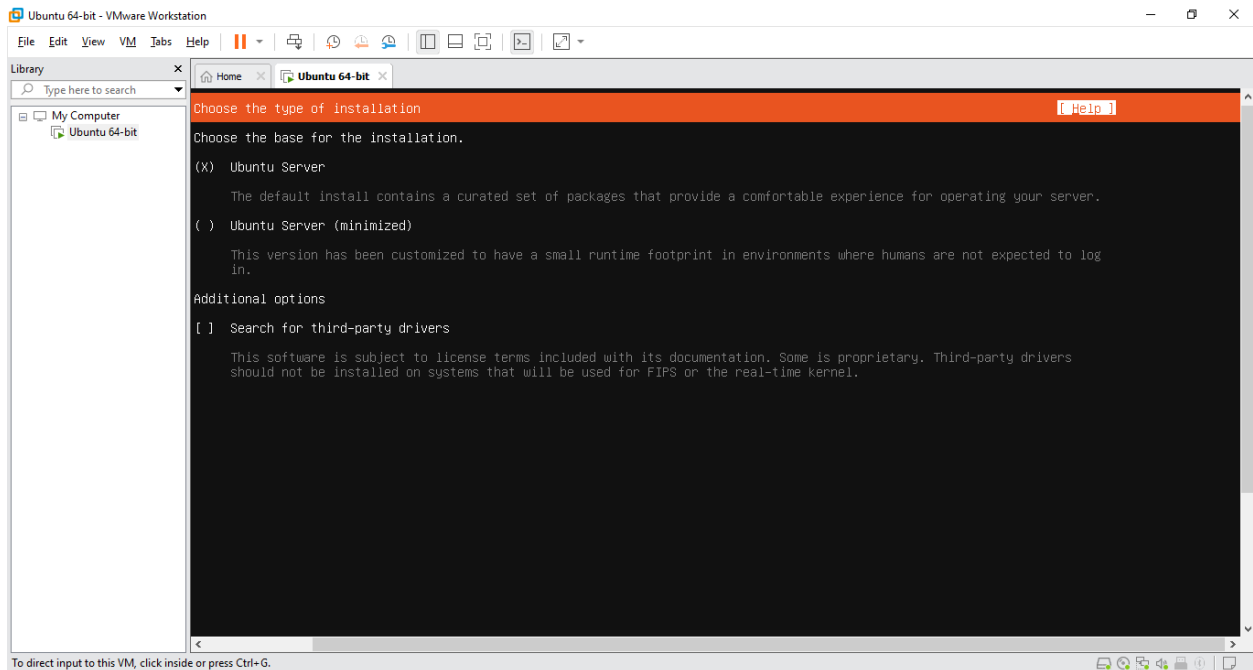


Choosing primary keyboard layout as English (US).



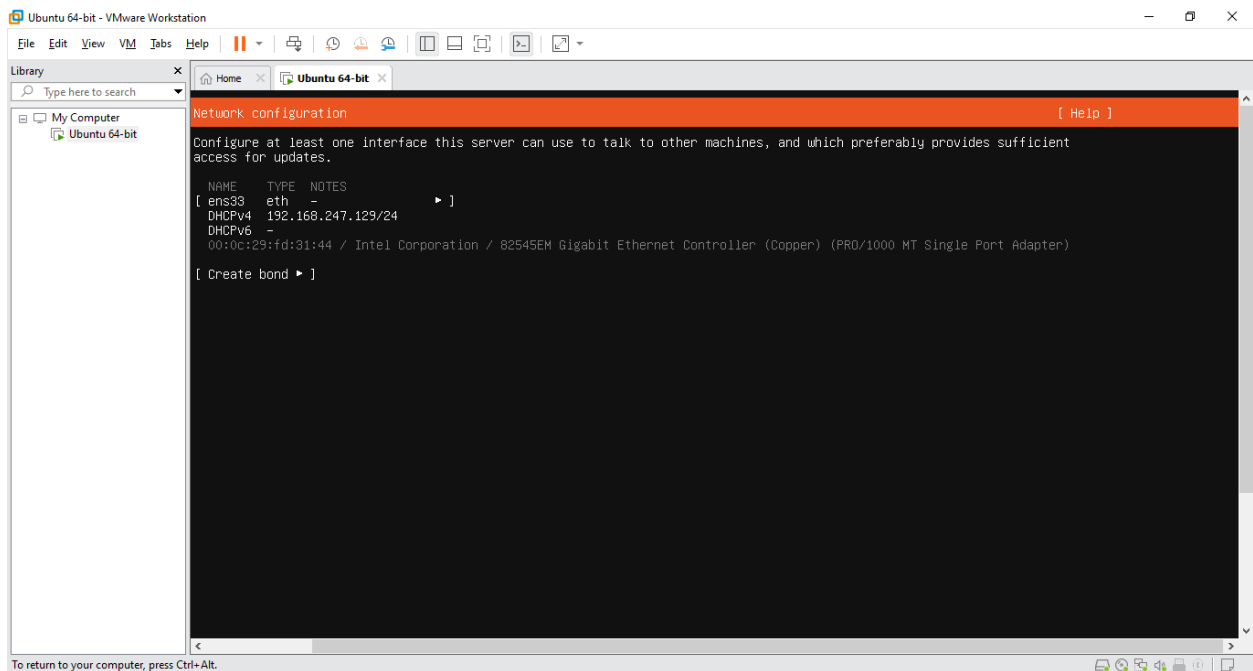
Confirming keyboard variant selection as English (US).

Task 7: Choose the Ubuntu Server Installation Type

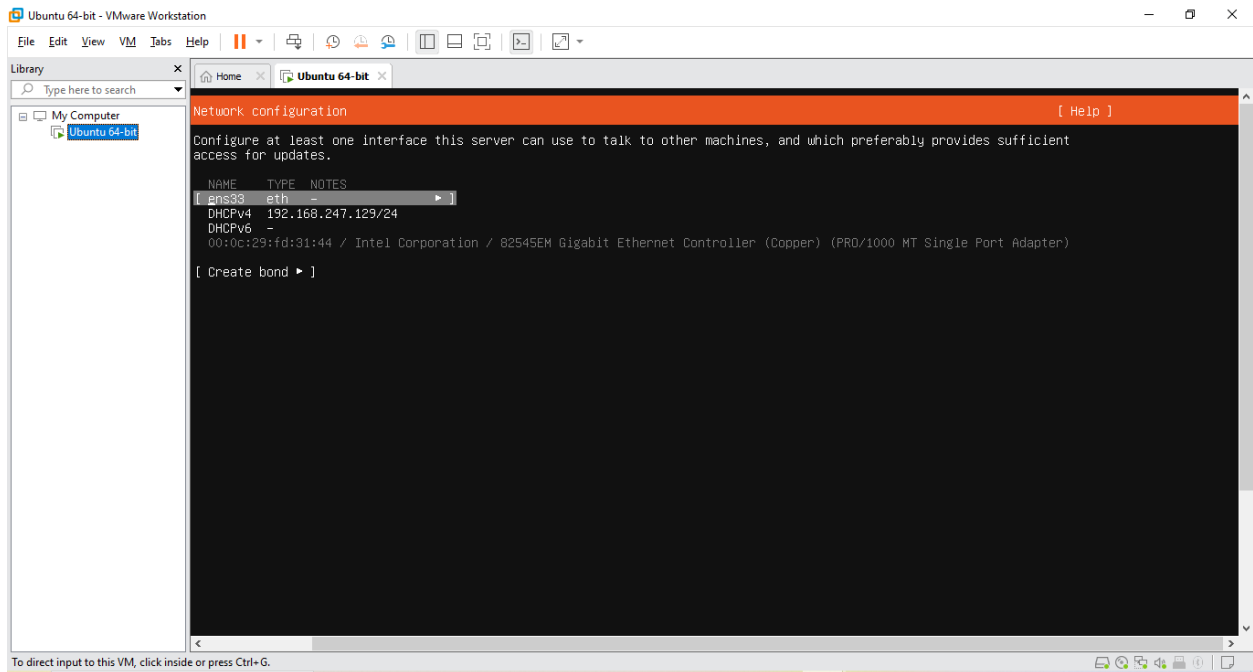


Selecting standard Ubuntu Server installation option.

Task 8: Configure the Network

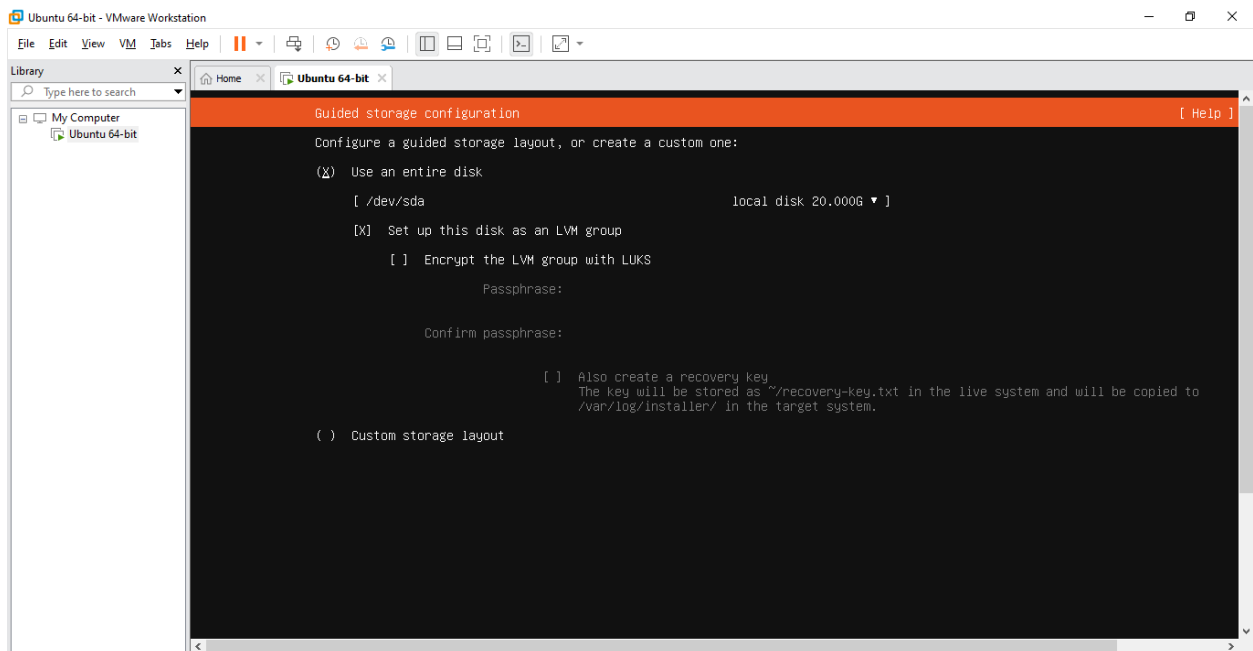


Automatic DHCP network IP assignment on interface ens33

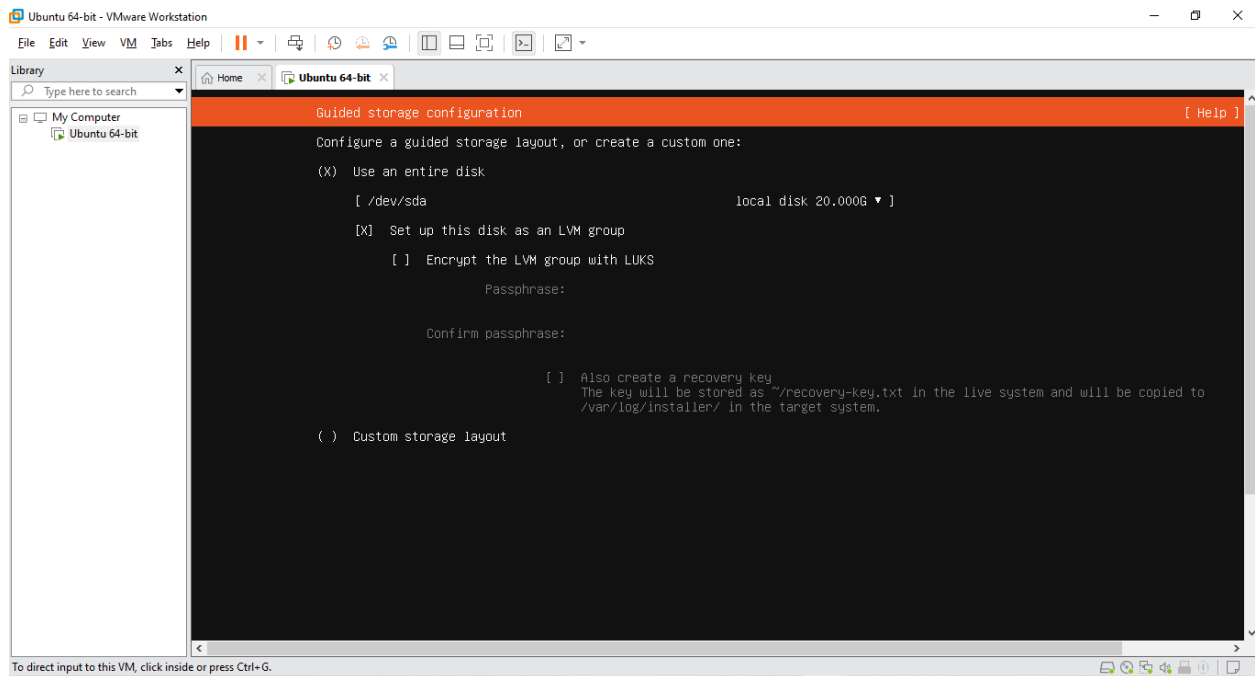


Active Ethernet connection confirmation.

Task 9: Configure Storage

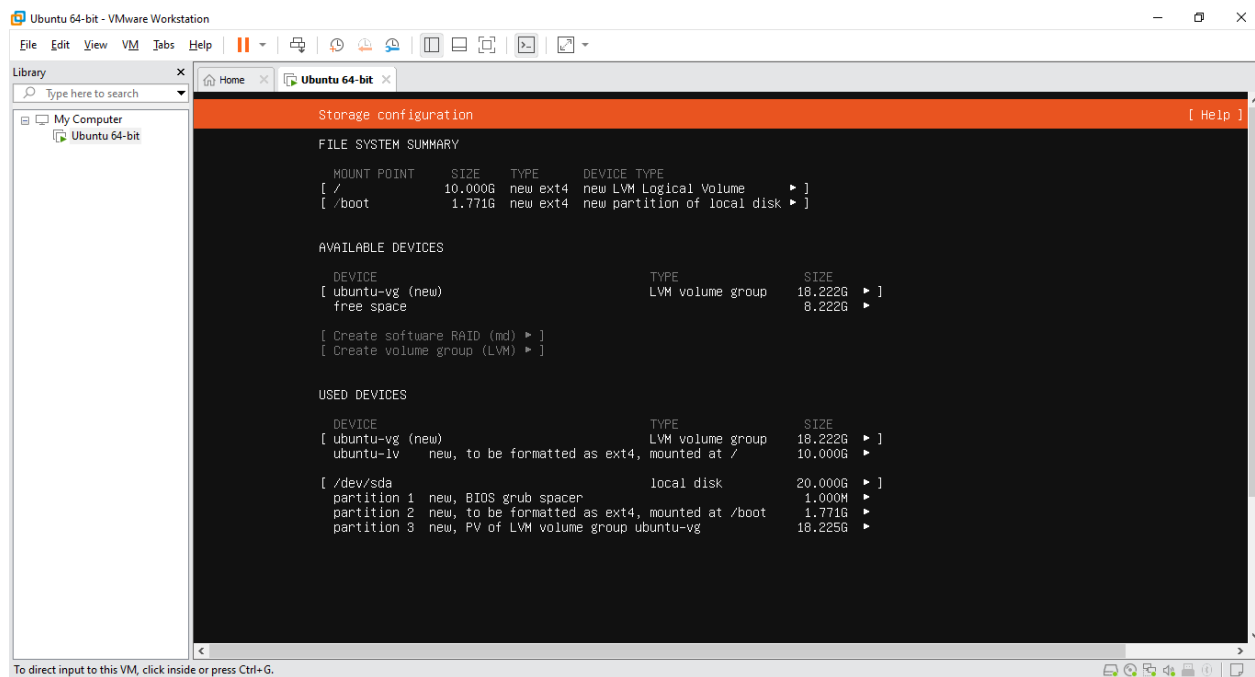


Configuring storage layout using entire 20GB disk with LVM group enabled.

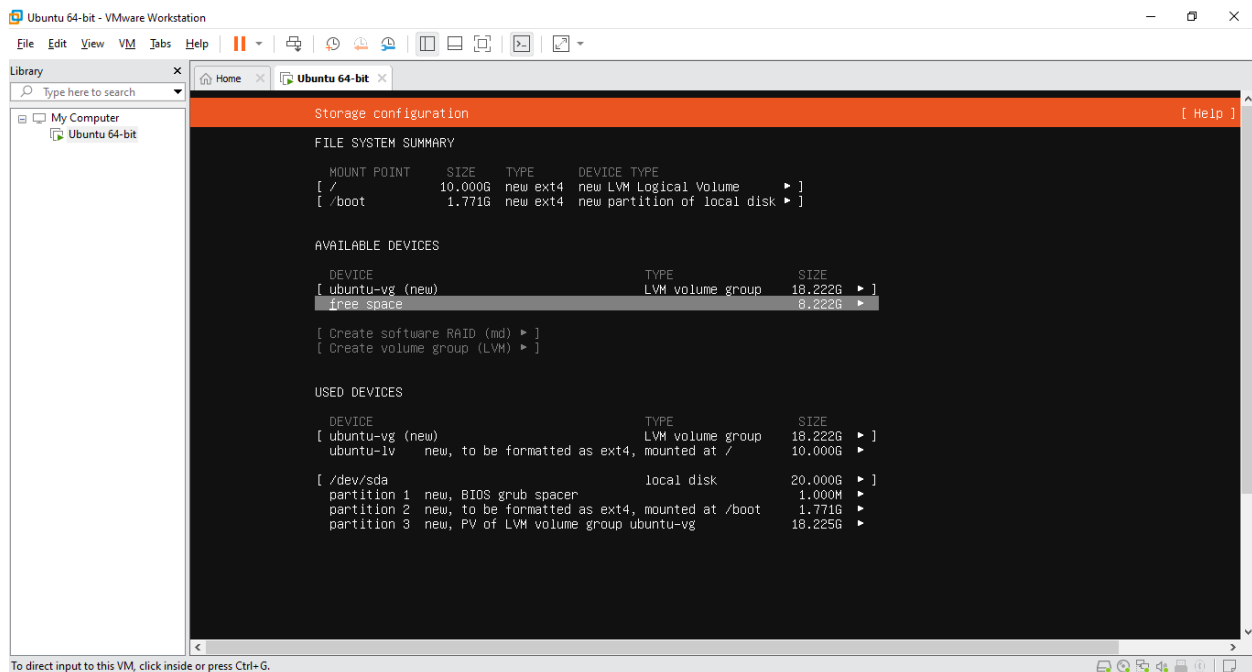


Finalized guided storage selection before filesystem generation.

Task 10: Select the Installation Disk

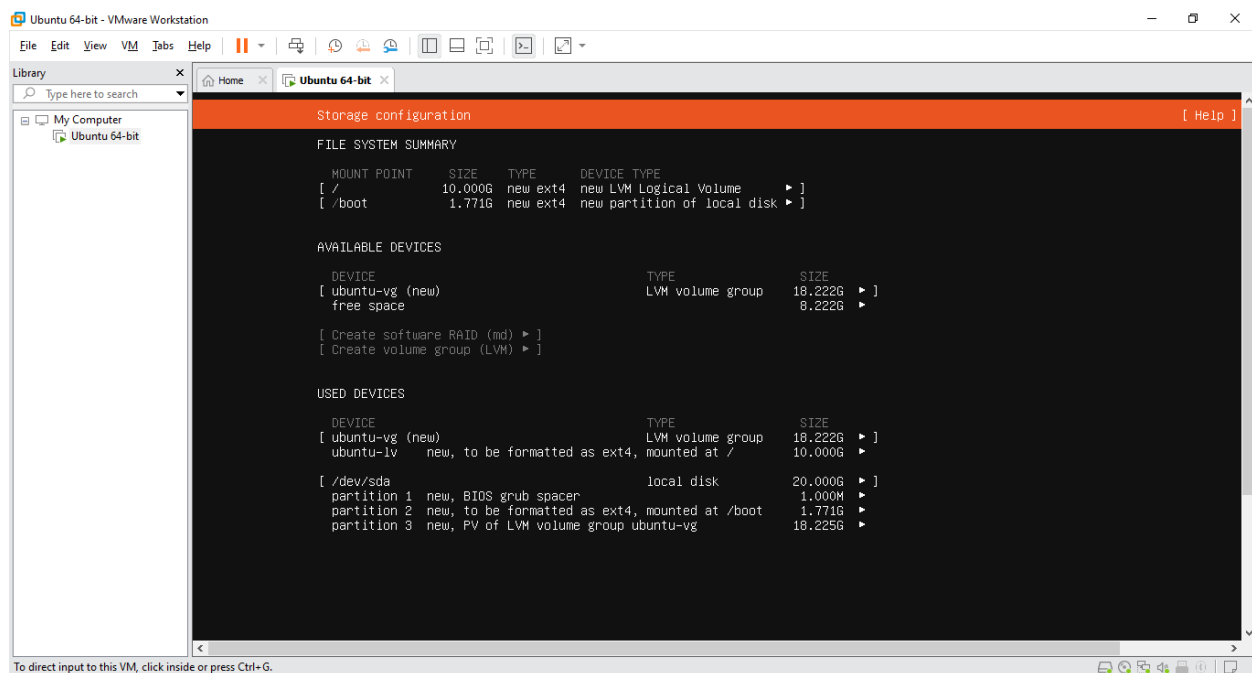


Summary view of mount points, root partition (/), and /boot configuration.



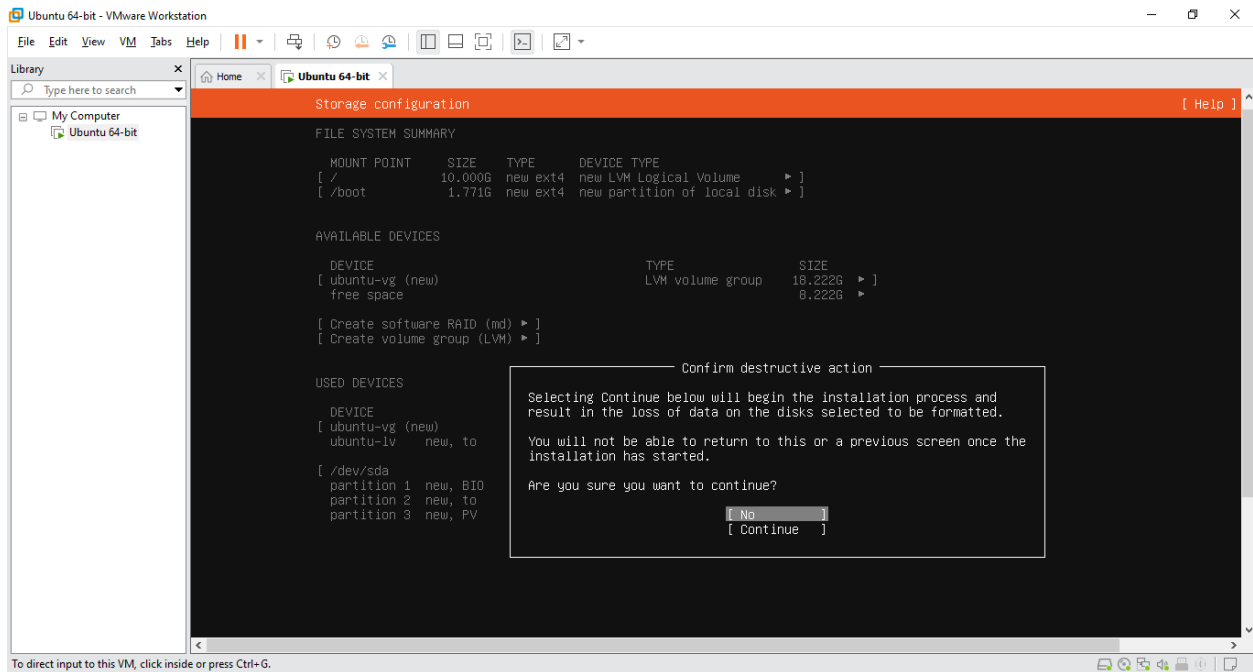
Disk partition allocation map showing LVM volume group storage breakdown.

Task 11: Review the Partitions

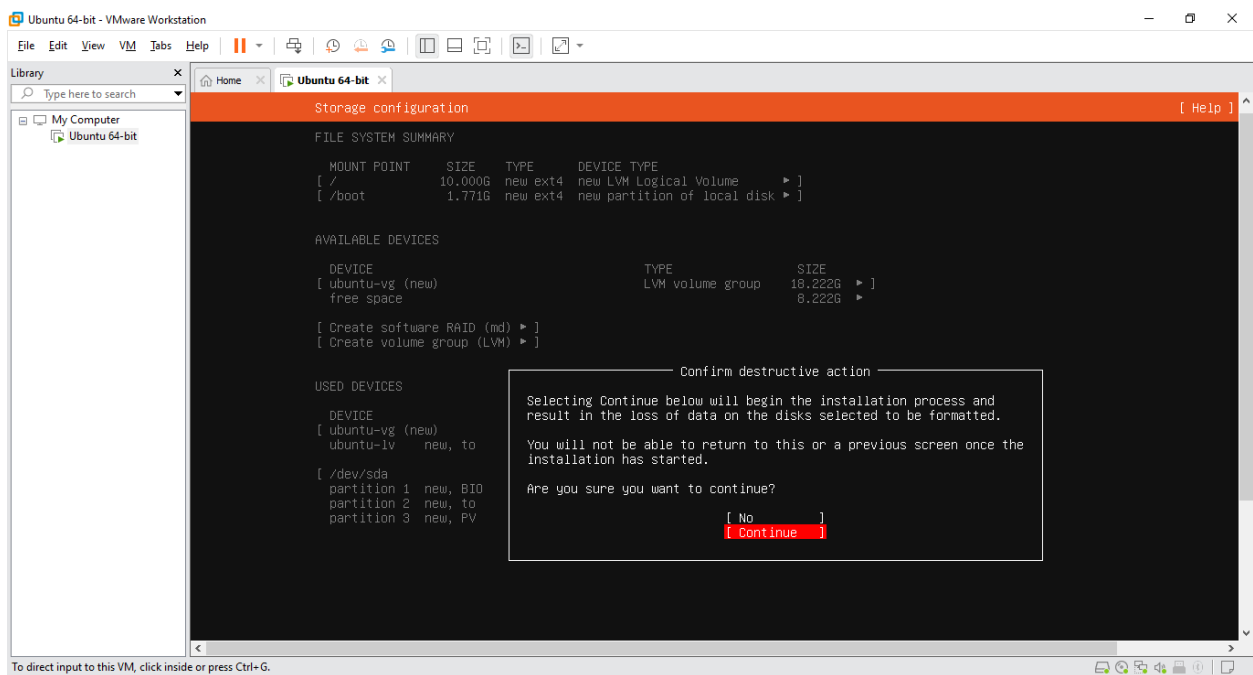


Final review of device mappings, LVM logical volumes, and ext4 formatting.

Task 12: Confirm the Storage Changes

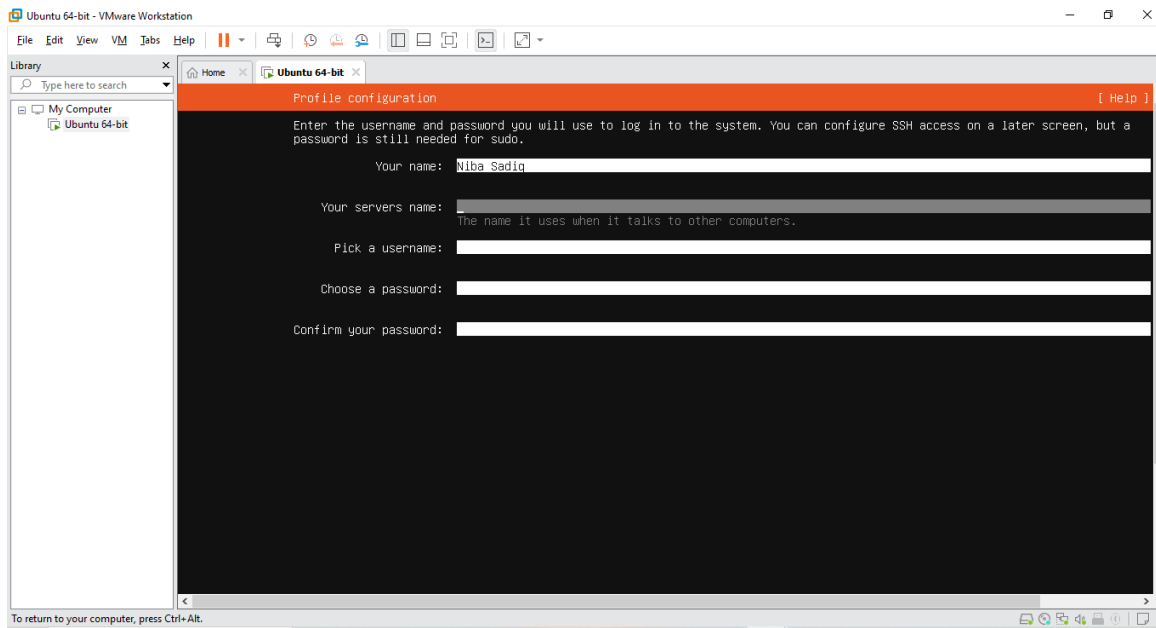


Confirmation pop-up warning about disk formatting.

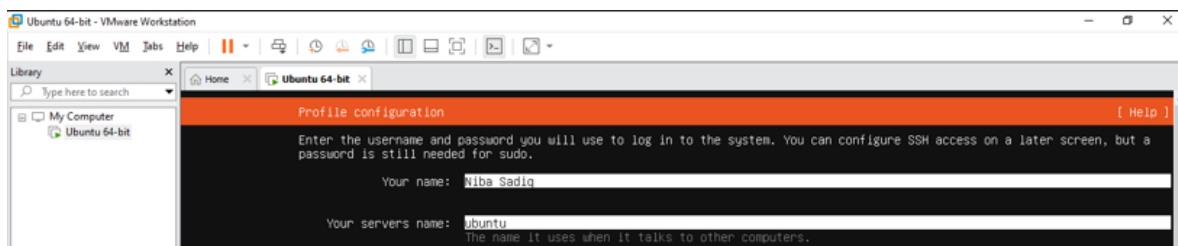


Selecting Continue to execute disk partitioning and formatting.

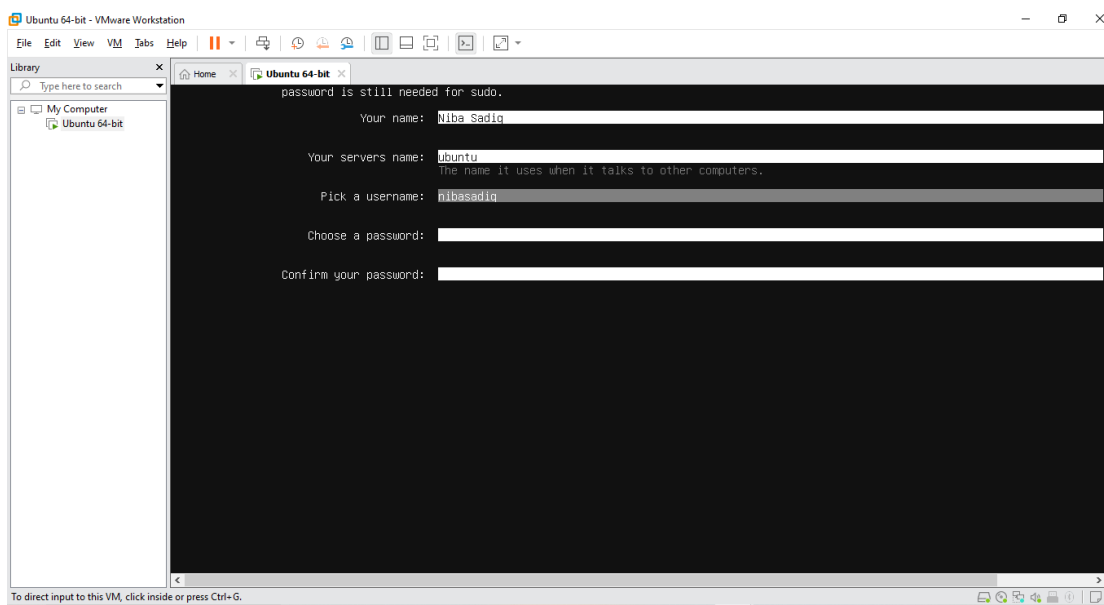
Task 13: Configure the Ubuntu Profile



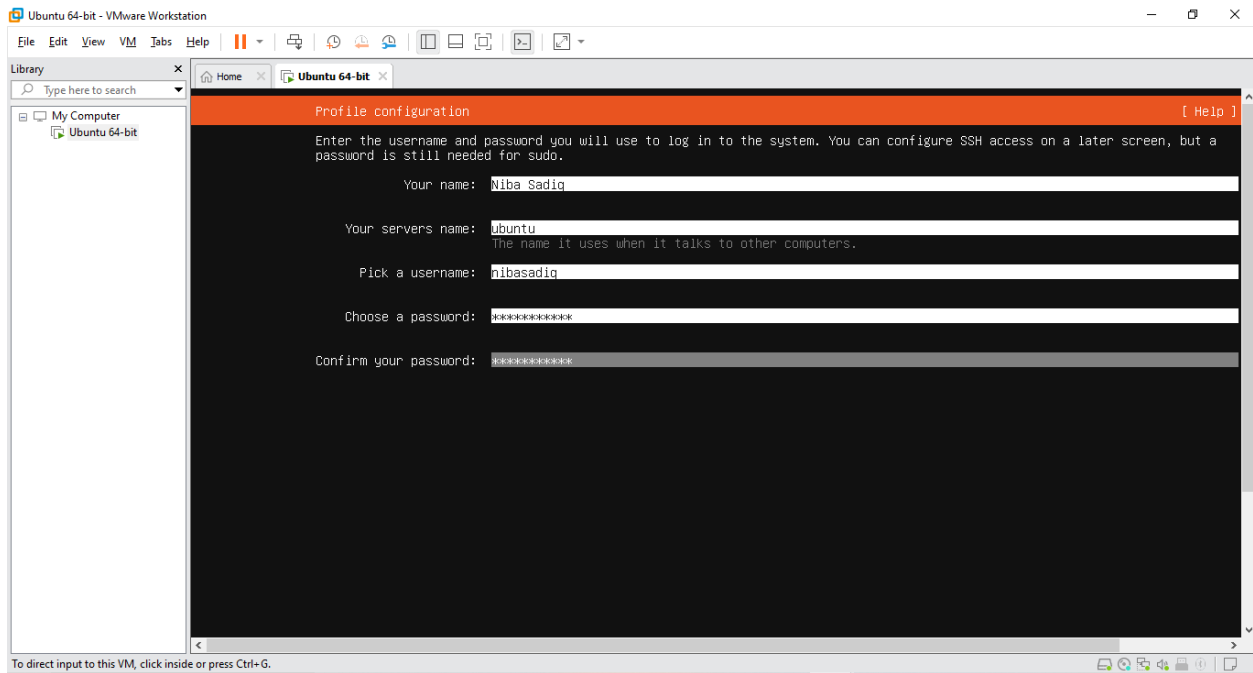
Real name field populated.



Setting server hostname to ubuntu.

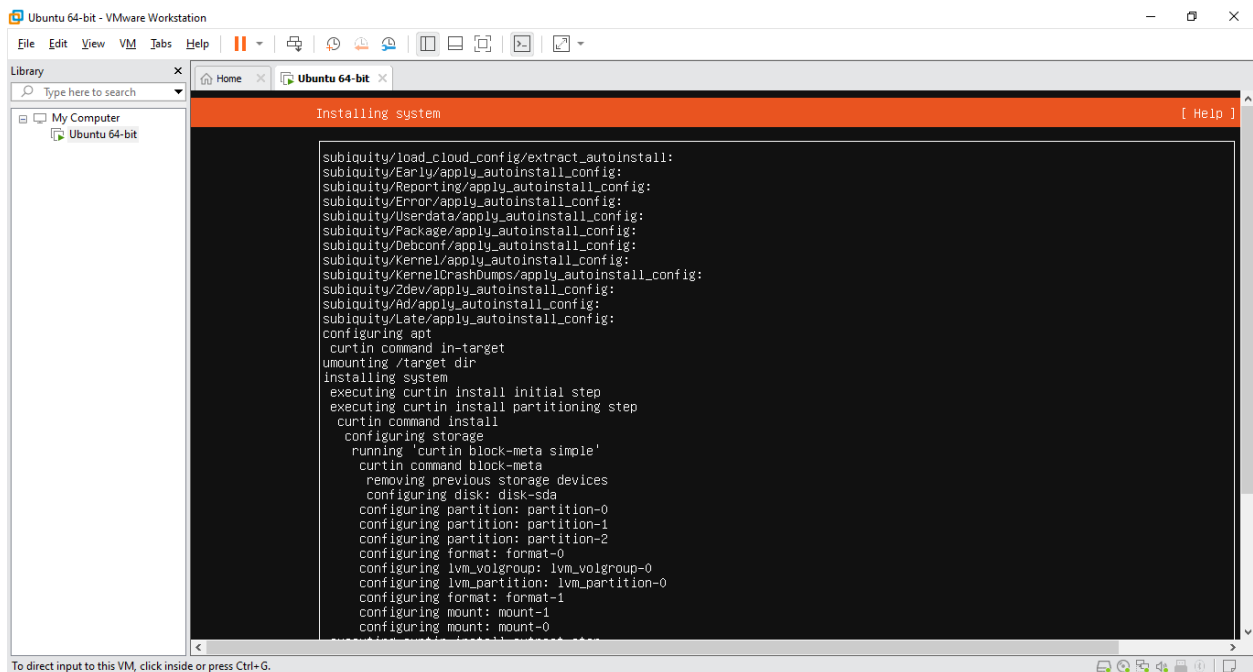


Configuring user account login name as nibasadiq.



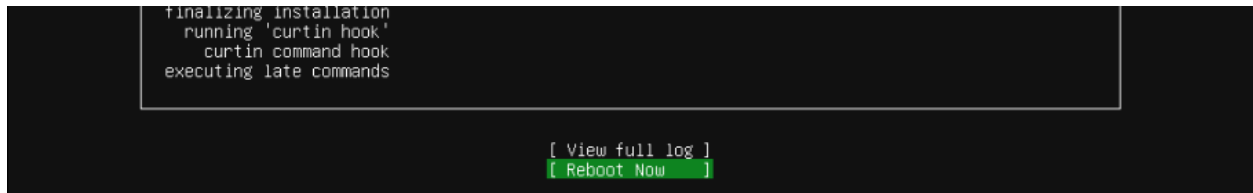
Entering and confirming user account password.

Task 14: Complete the Software Installation

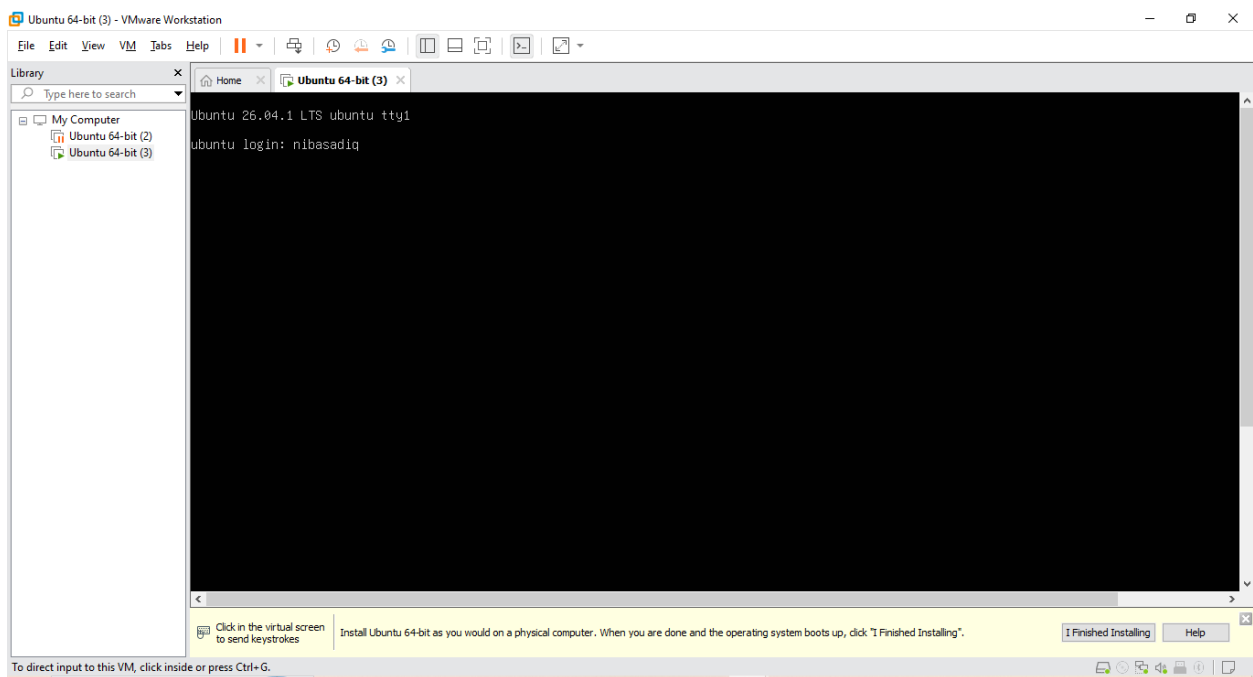


Base system installation log showing automated package deployment and storage setup.

Task 15: Reboot and Verify the Ubuntu Identity



Selecting "Reboot Now" after installation finishes.



Ubuntu Server login prompt displaying hostname and login field

The screenshot shows the terminal output of the cloud-init process during the first boot of Ubuntu 64-bit. The process generates an SSH host key pair (ed25519) and displays the key fingerprint. The output is as follows:

```
52.7880711 cloud-init[1201]: SHA256:2SzNw1C6tdxkwy4IT6Ts02F5+h8c51N7UuU+33jV61 root@ubuntu
52.7929901 cloud-init[1201]: The key's randomart image is:
52.7957921 cloud-init[1201]: +---[ECDSA 256]----+
52.7979351 cloud-init[1201]: | . . . |
52.7993871 cloud-init[1201]: | . . . o + |
52.8019601 cloud-init[1201]: | .+... + * |
52.8040961 cloud-init[1201]: | * ooo. = * |
52.8084481 cloud-init[1201]: | *.=. . S .o |
52.8105681 cloud-init[1201]: | + + .o o ..... |
52.8135051 cloud-init[1201]: | .= o. o. + .o. |
52.8159131 cloud-init[1201]: | . o = ..o ..E. oo |
52.8184931 cloud-init[1201]: | . +. ..O.= |
52.8208311 cloud-init[1201]: +---[SHA256]-----+
52.8220291 cloud-init[1201]: Generating public/private ed25519 key pair.
52.8259521 cloud-init[1201]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
52.8286441 cloud-init[1201]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
52.8306901 cloud-init[1201]: The key fingerprint is:
52.8327301 cloud-init[1201]: SHA256:5M82/mL+o0gyB68kc59s7NXJCMoBqsKFJrLLXv8ZbDM root@ubuntu
52.8344781 cloud-init[1201]: The key's randomart image is:
52.8367241 cloud-init[1201]: +---[ED25519 256]---+
52.8396331 cloud-init[1201]: |
52.8428721 cloud-init[1201]: |
52.8448671 cloud-init[1201]: | .
52.8488281 cloud-init[1201]: | ... o
52.8502751 cloud-init[1201]: | +.o S
52.8519411 cloud-init[1201]: | +o.o +. +
52.8664261 cloud-init[1201]: | *.o.E==+ *
52.8687571 cloud-init[1201]: | o=.o+Bo.++o
52.8741591 cloud-init[1201]: | oo.+Bo+o.oo+o
52.8786451 cloud-init[1201]: +---[SHA256]-----+

ubuntu login:
ubuntu login: nibasadiq
Password:

Click in the virtual screen to send keystrokes | Install Ubuntu 64-bit as you would on a physical computer. When you are done and the operating system boots up, click "I Finished Installing".
I Finished Installing | Help
```

System cloud-init process generating SSH host keys on first boot.

The screenshot shows the system status and login prompt after the cloud-init process has completed. The output is as follows:

```
System load: 0.17          Processes:      216
Usage of /:  20.9% of 9.75GB Users logged in: 0
Memory usage: 6%          IPv4 address for ens33: 192.168.247.133
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

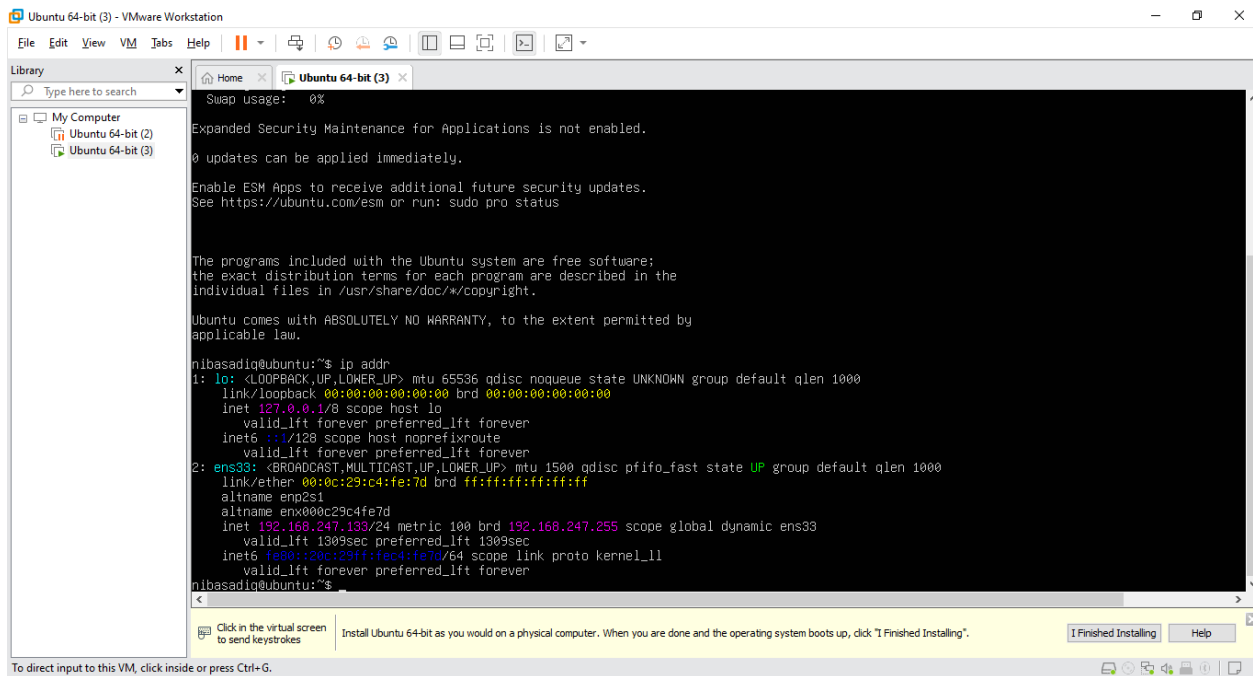
The programs included with the Ubuntu system are free software:
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

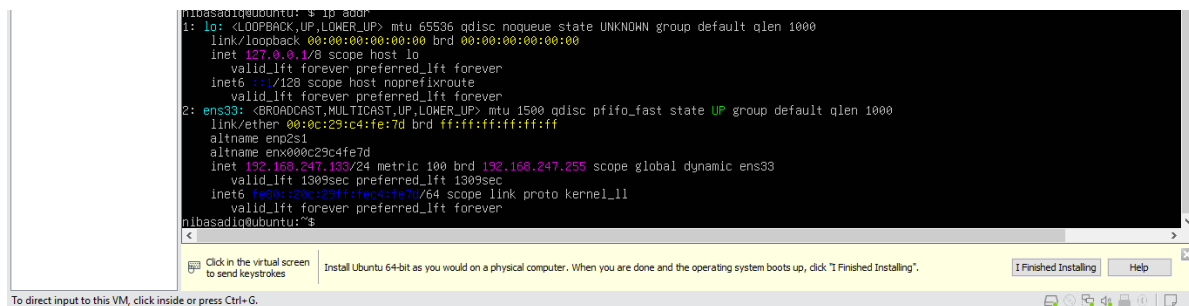
nibasadiq@ubuntu:~$ _
```

Logged into bash prompt nibasadiq@ubuntu:~\$.

Task 16: Find the Ubuntu Server IP Address



Output of ip addr showing network interfaces and assigned IPv4 address 192.168.247.133 on ens33.



Zoomed view verifying subnet mask and dynamic IP lease.

Task 17: Connect to Ubuntu Server from Windows

```
nibasadiq@ubuntu:~$ ssh nibasadiq@192.168.247.133
The authenticity of host '192.168.247.133 (192.168.247.133)' can't be established.
ED25519 key fingerprint is: SHA256:5M82/mL+oOgYB68Kc59s7NXJCMoBqsKFJrLLXv8ZbCM
This key is not known by any other names.
```

Windows command prompt executing SSH command.

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.247.133' (ED25519) to the list of known hosts.
```

Key fingerprint verification prompt.

```
nibasadiq@192.168.247.133's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri 18 Sept 07:22:21 UTC 2026

System load:  0.25               Processes:            228
Usage of /:   30.6% of 9.75GB    Users logged in:     0
Memory usage: 8%                IPv4 address for ens33: 192.168.247.133
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

109 updates can be applied immediately.
96 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Established remote SSH session displaying system status and system load.

```
nibasadiq@ubuntu:~$
```

Active remote SSH shell session.